

Проектирование и администрирование серверной инфраструктуры веб-сервисов

Московский колледж бизнес-технологий
Москва, 2022

Цели

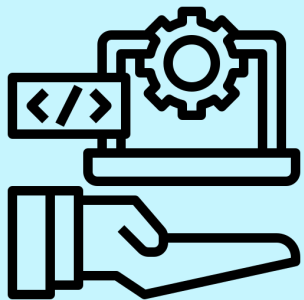
- Выбор и конфигурация ПО
- Развертывание сервиса
- Обеспечение доступа к сервису
- Защита сервиса

Задачи

- Сравнить и выбрать технологии для развертывания
- Выбрать ОС
- Настроить и защитить инфраструктуру

Анализ существующих типов платформ для развертывания веб-приложений и веб- сервисов

Веб – {сервисы, приложения}



Примеры типов веб-сервисов

- XML-RPC
- JSON-RPC
- SOAP
- REST

Классификация платформ

- In-House
- Colocation
- Managed Hosting
 - ↳ Dedicated Server
 - ↳ Virtual Private Server
 - ↳ Cloud Computing
 - └ IaaS
 - └ PaaS
 - └ SaaS
 - └ Serverless Computing

Мониторинг ресурсов

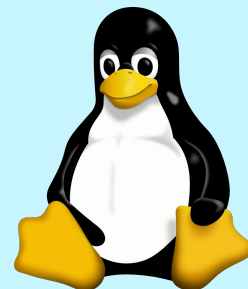
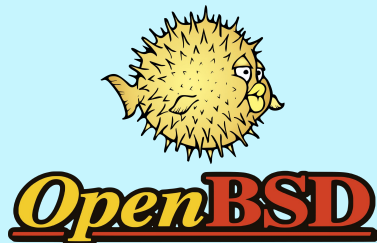
CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS
1c6ae4c0b434	www	0.00%	26.9MiB / 1.937GiB	1.36%	5.01MB / 191MB	10.9MB / 8.19kB	2
ff8cb9d80f76	sfl	0.00%	162.3MiB / 1.937GiB	8.18%	63.3MB / 3.54MB	0B / 0B	2

Сравнение стоимости услуг

Провайдер	Тип платформы	Стоимость (руб. /месяц)
Microsoft Azure	Cloud Computing	~4 938
Google Cloud	Cloud Computing	~12 228
Heroku	PaaS	2 550
REG.RU	VPS	~370
AdminVPS	VPS	809

Практическое развертывание программного обеспечения сервиса

Операционные системы



Ubuntu | Debian



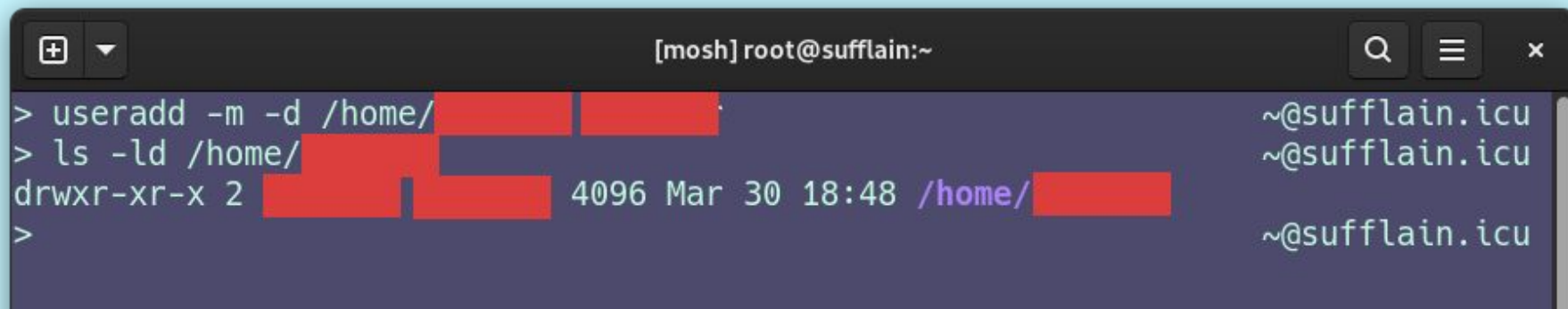
Что необходимо

- SSH
- Баннеры
- Межсетевой экран
- IPS
- Docker (+ развернуть контейнеры)

Выбор ПО

- SSH - openssh
- Mosh
- IPS - fail2ban
- Firewall - ufw
- БД - Firebase Realtime

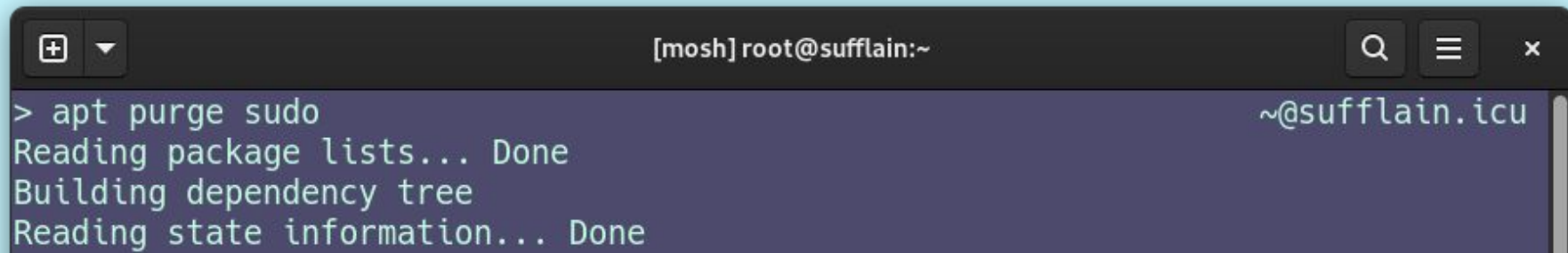
Создание пользователя



A terminal window with a dark background and light text. The title bar shows "[mosh] root@sufflain:~". The terminal content shows the execution of two commands: "useradd -m -d /home/[redacted]" and "ls -ld /home/[redacted]". The output of the second command shows a directory with permissions "drwxr-xr-x", owned by "2 [redacted]", size "4096", date "Mar 30 18:48", and path "/home/[redacted]". The prompt ">" is visible at the end of each line. On the right side of the terminal, the prompt "~@sufflain.icu" is repeated three times, corresponding to the three lines of output.

```
[mosh] root@sufflain:~  
> useradd -m -d /home/[redacted] . ~@sufflain.icu  
> ls -ld /home/[redacted] ~@sufflain.icu  
drwxr-xr-x 2 [redacted] 4096 Mar 30 18:48 /home/[redacted] ~@sufflain.icu  
>
```

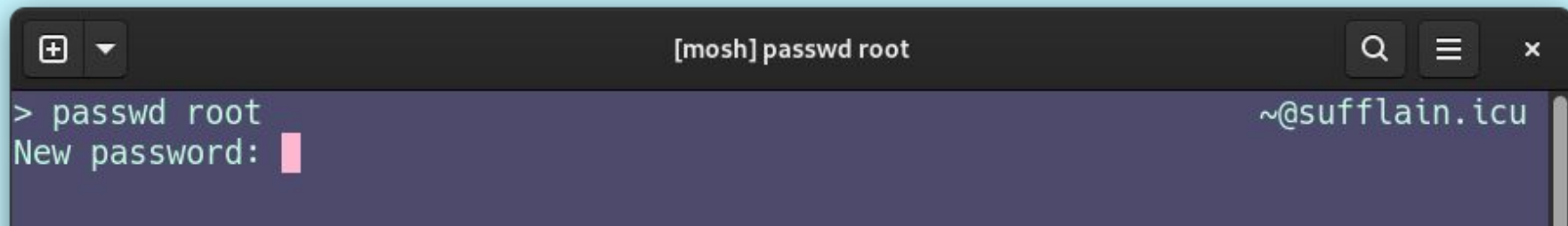
Полное удаление sudo



```
[mosh] root@sufflain:~  
> apt purge sudo  
Reading package lists... Done  
Building dependency tree  
Reading state information... Done
```

The image shows a terminal window with a dark theme. The title bar at the top contains a window control icon (a square with a plus sign and a downward arrow), the text "[mosh] root@sufflain:~", and standard window controls (a magnifying glass for search, a hamburger menu, and a close button 'x'). The terminal content shows the execution of the command 'apt purge sudo'. The output consists of four lines: the command prompt followed by the command, and then three status messages: 'Reading package lists... Done', 'Building dependency tree', and 'Reading state information... Done'. The prompt '~@sufflain.icu' is visible on the right side of the terminal window.

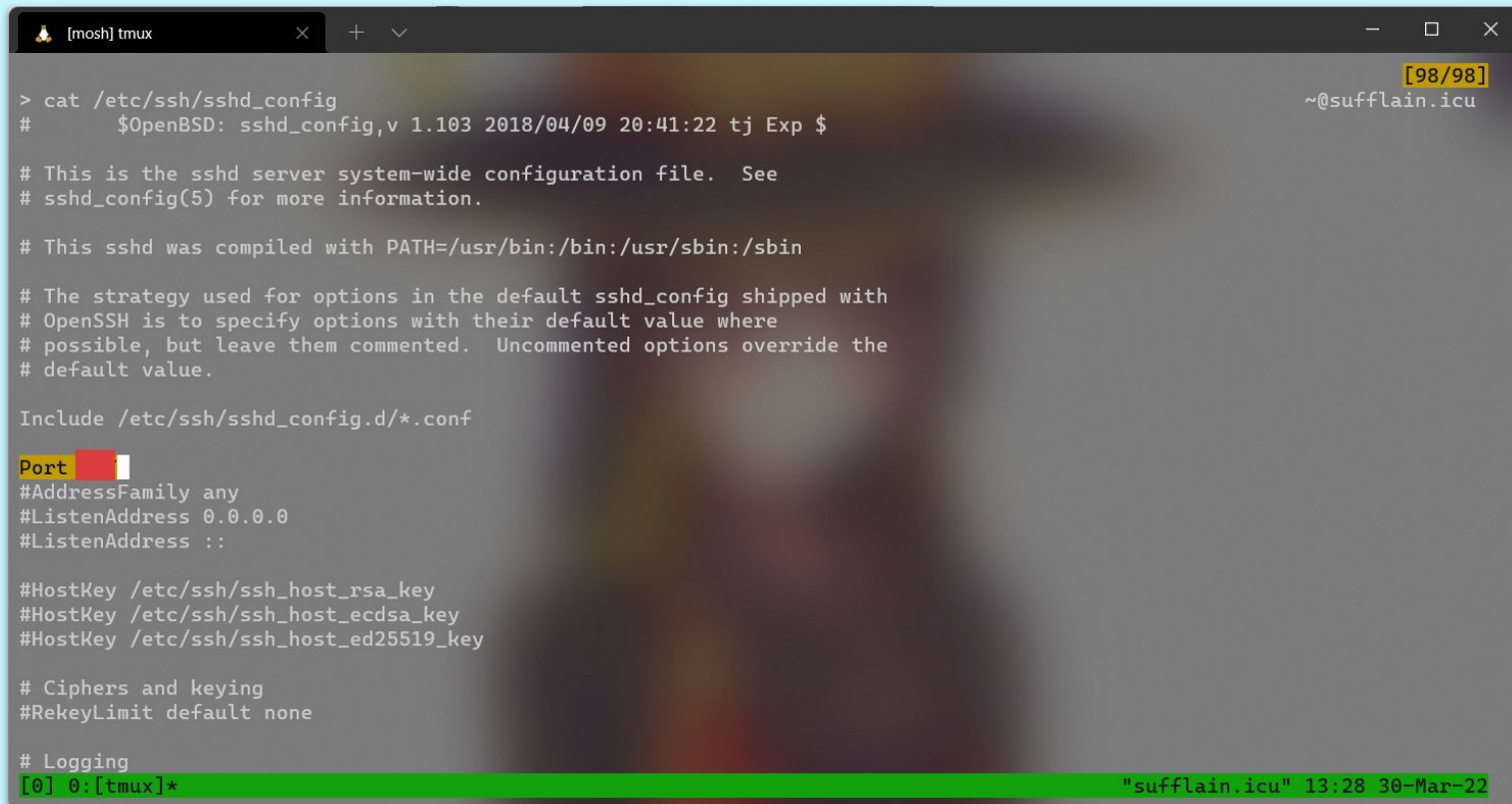
Изменение пароля пользователя root



```
[mosh] passwd root
> passwd root
New password: 
```

The image shows a terminal window with a dark background. The title bar at the top contains a window control icon (a square with a plus sign and a downward arrow), the text "[mosh] passwd root", and three icons on the right: a magnifying glass, a hamburger menu, and a close button (an 'x'). The terminal content shows a prompt ">" followed by the command "passwd root". Below this, the text "New password:" is displayed, followed by a red rectangular cursor. The username "~@sufflain.icu" is visible in the top right corner of the terminal area.

Порт SSH



A terminal window titled "[mosh] tmux" displays the contents of the file `/etc/ssh/sshd_config`. The window has standard window controls (close, maximize, zoom) and a status bar at the bottom. The status bar on the right shows "[98/98]" and "~@sufflain.icu". The status bar at the bottom is green and shows "[0] 0:[tmux]*" on the left and "\"sufflain.icu\" 13:28 30-Mar-22" on the right.

```
> cat /etc/ssh/sshd_config
#      $OpenBSD: sshd_config,v 1.103 2018/04/09 20:41:22 tj Exp $

# This is the sshd server system-wide configuration file.  See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/bin:/bin:/usr/sbin:/sbin

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented.  Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

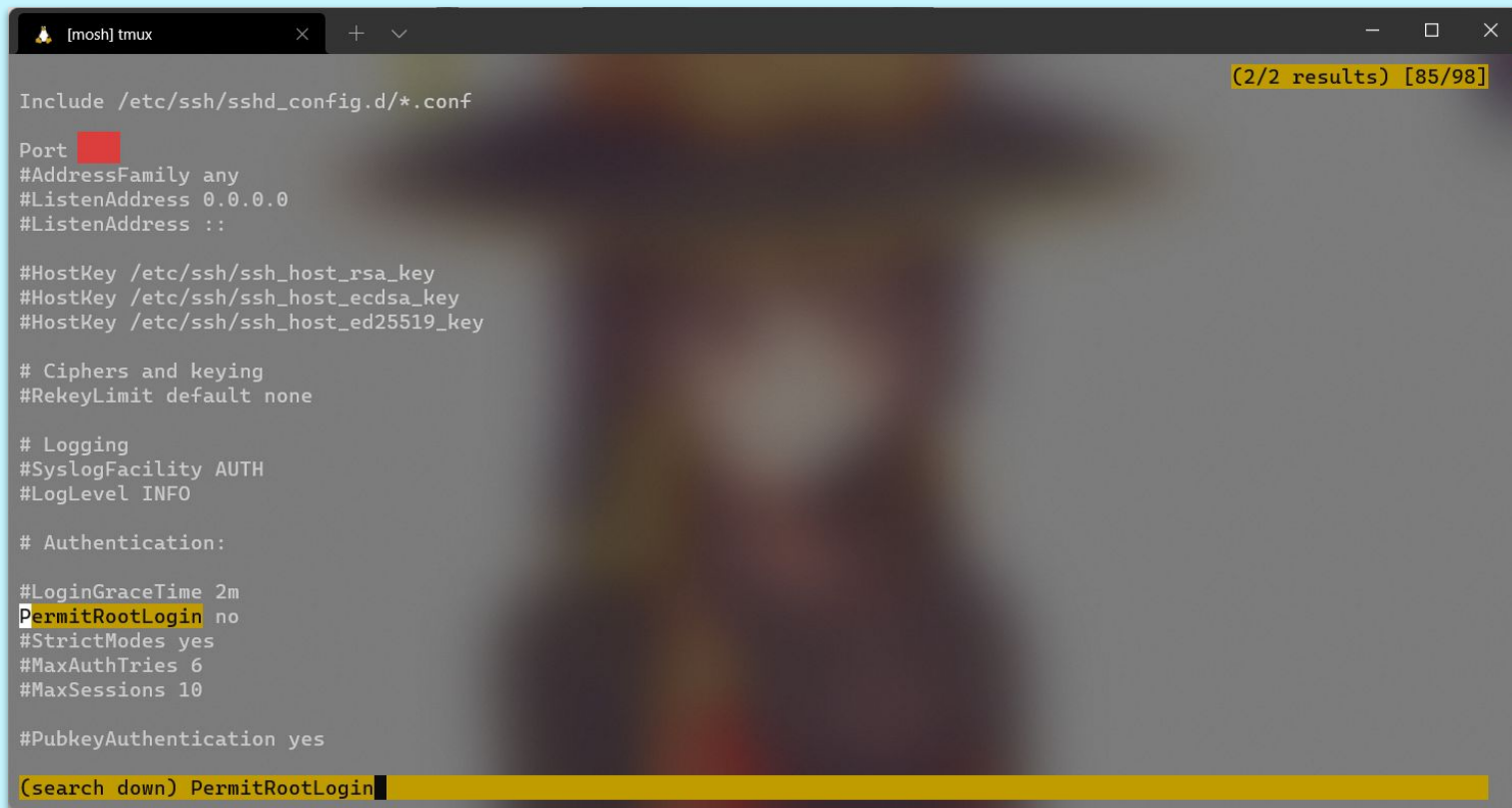
Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
```

Запрет на вход в root через SSH



```
[mosh] tmux
(2/2 results) [85/98]
Include /etc/ssh/sshd_config.d/*.conf

Port [REDACTED]
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

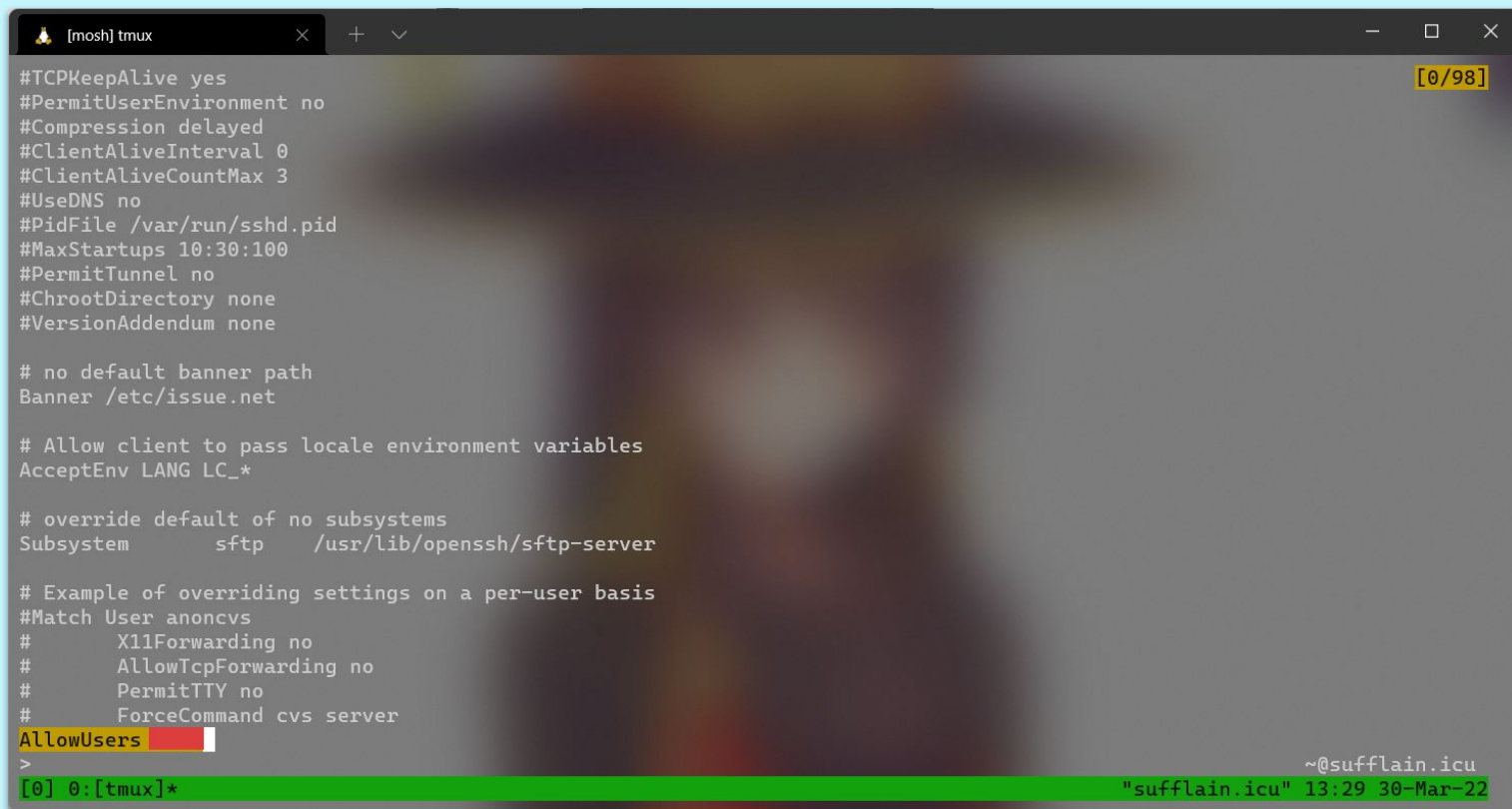
# Authentication:

#LoginGraceTime 2m
PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

(search down) PermitRootLogin
```

Разрешение входа через SSH в созданный аккаунт



The image shows a terminal window titled "[mosh] tmux". The terminal displays the contents of the `/etc/ssh/sshd_config` file, which includes various SSH server settings. The configuration is as follows:

```
#TCPKeepAlive yes
#PermitUserEnvironment no
#Compression delayed
#ClientAliveInterval 0
#ClientAliveCountMax 3
#UseDNS no
#PidFile /var/run/sshd.pid
#MaxStartups 10:30:100
#PermitTunnel no
#ChrootDirectory none
#VersionAddendum none

# no default banner path
Banner /etc/issue.net

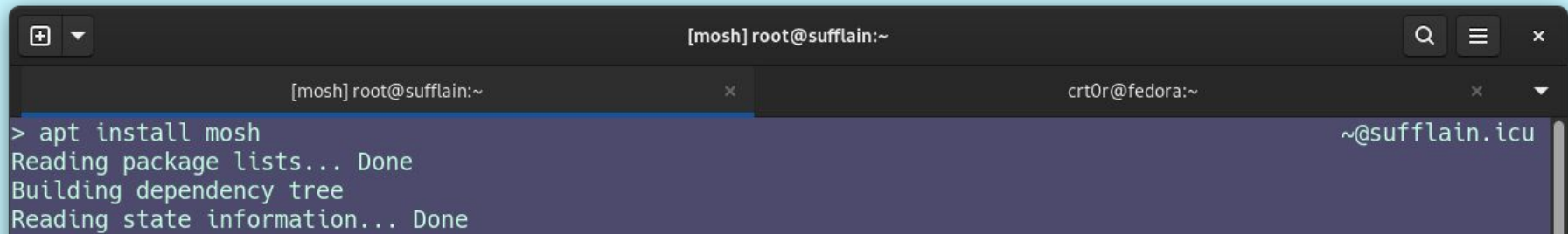
# Allow client to pass locale environment variables
AcceptEnv LANG LC_*

# override default of no subsystems
Subsystem          sftp          /usr/lib/openssh/sftp-server

# Example of overriding settings on a per-user basis
#Match User anoncvs
#      X11Forwarding no
#      AllowTcpForwarding no
#      PermitTTY no
#      ForceCommand cvs server
AllowUsers
```

The terminal shows a red cursor at the end of the `AllowUsers` line. Below this, a green status bar indicates a successful login: `[0] 0:[tmux]*` on the left and `~@sufflain.icu "sufflain.icu" 13:29 30-Mar-22` on the right.

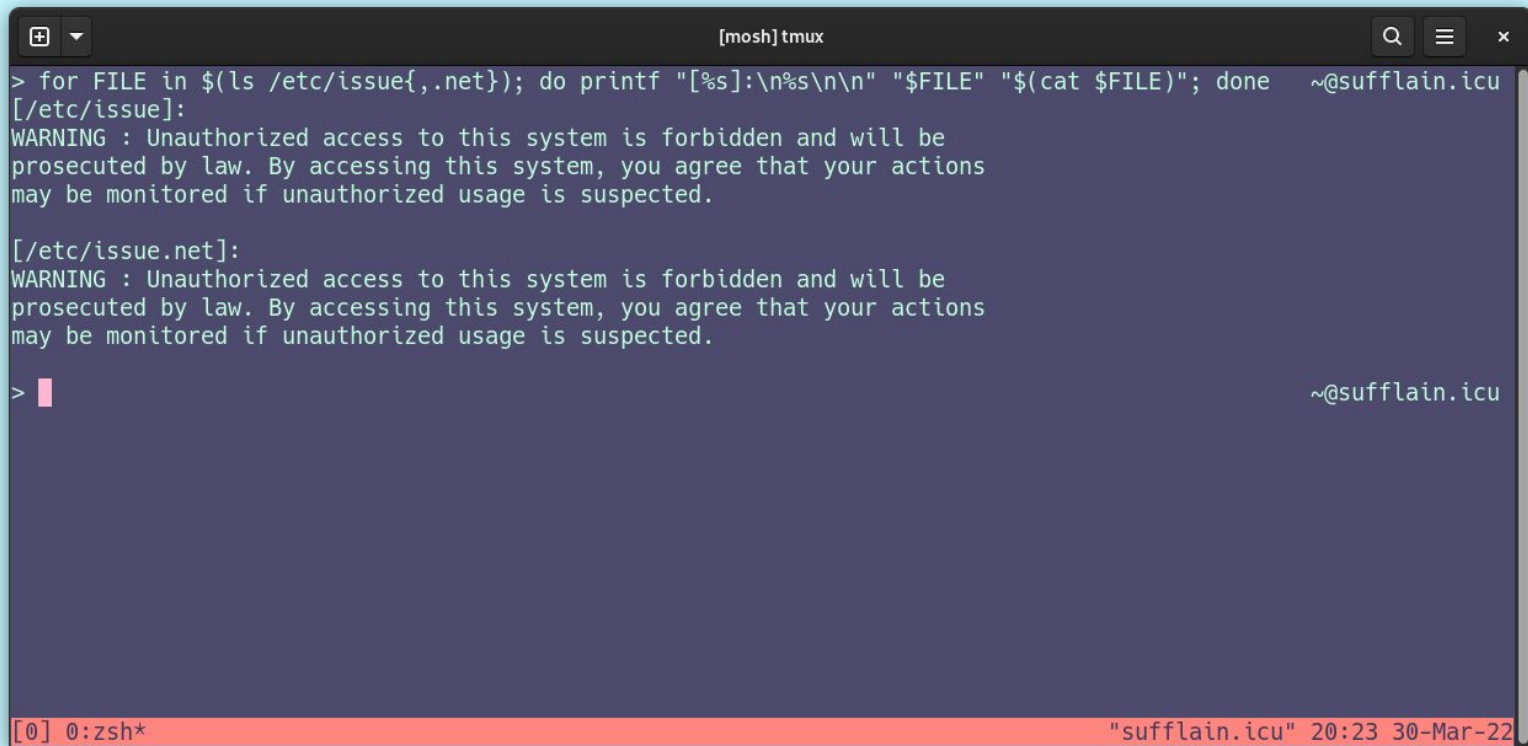
Установка Mosh



A terminal window with a dark theme. The title bar shows '[mosh] root@sufflain:~'. The terminal content shows the command 'apt install mosh' being executed, followed by the output: 'Reading package lists... Done', 'Building dependency tree', and 'Reading state information... Done'. The prompt '~@sufflain.icu' is visible on the right side of the terminal.

```
[mosh] root@sufflain:~  
> apt install mosh  
Reading package lists... Done  
Building dependency tree  
Reading state information... Done  
~@sufflain.icu
```

Баннеры



```
[mosh] tmux
> for FILE in $(ls /etc/issue{,.net}); do printf "[%s]:\n%s\n\n" "$FILE" "$(cat $FILE)"; done ~@sufflain.icu
[/etc/issue]:
WARNING : Unauthorized access to this system is forbidden and will be
prosecuted by law. By accessing this system, you agree that your actions
may be monitored if unauthorized usage is suspected.

[/etc/issue.net]:
WARNING : Unauthorized access to this system is forbidden and will be
prosecuted by law. By accessing this system, you agree that your actions
may be monitored if unauthorized usage is suspected.

> █ ~@sufflain.icu
```

[0] 0:zsh* "sufflain.icu" 20:23 30-Mar-22

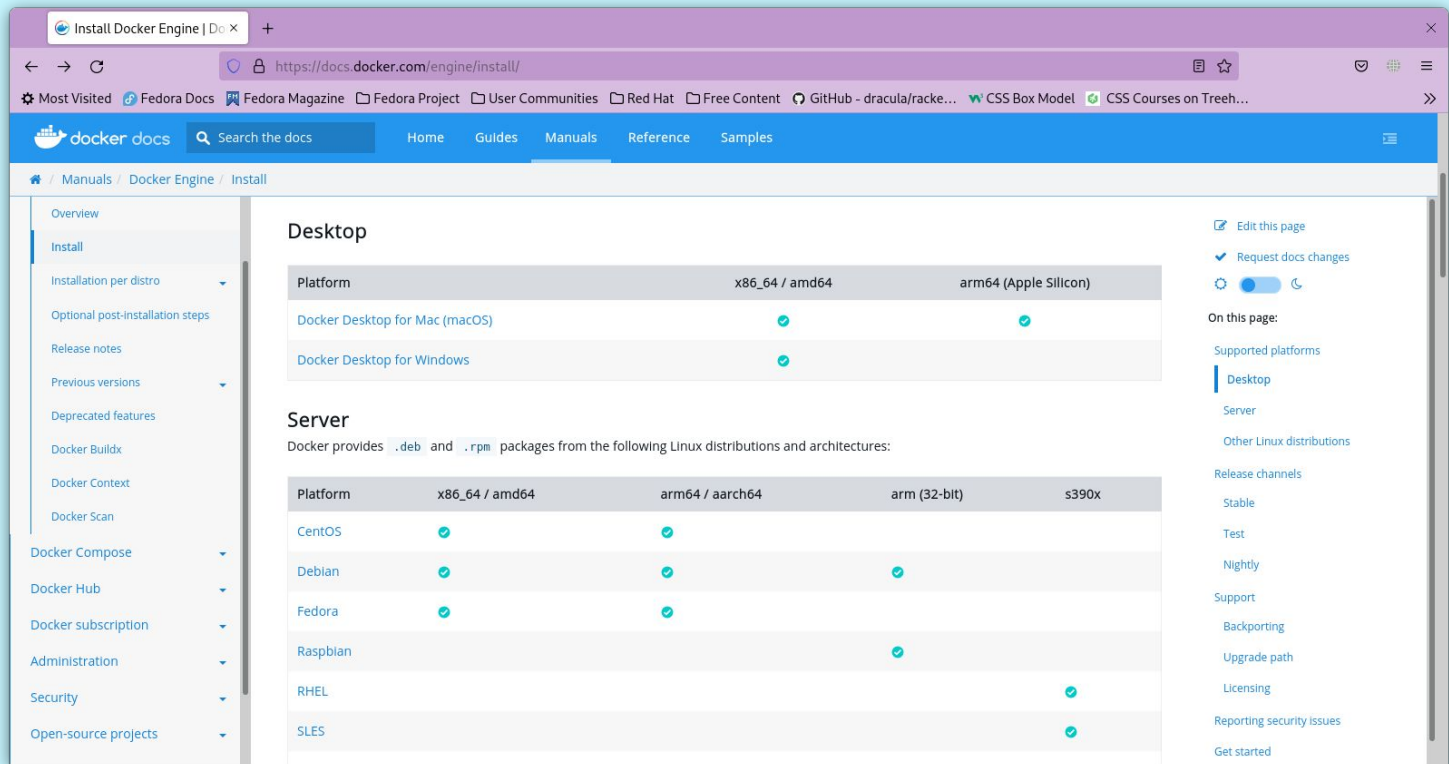
fail2ban

```
systemctl status fail2ban

● fail2ban.service - Fail2Ban Service
   Loaded: loaded (/lib/systemd/system/fail2ban.service; enabled; vendor preset: enabled)
   Active: active (running) since Sat 2022-03-12 08:38:02 MSK; 3 weeks 0 days ago
     Docs: man:fail2ban(1)
   Process: 450737 ExecReload=/usr/bin/fail2ban-client reload (code=exited, status=0/SUCCESS)
  Main PID: 506 (f2b/server)
    Tasks: 5 (limit: 2311)
   Memory: 15.6M
   CGroup: /system.slice/fail2ban.service
           └─506 /usr/bin/python3 /usr/bin/fail2ban-server -xf start

Mar 12 08:38:02 sufflain.icu systemd[1]: Starting Fail2Ban Service...
Mar 12 08:38:02 sufflain.icu systemd[1]: Started Fail2Ban Service.
Mar 12 08:38:03 sufflain.icu fail2ban-server[506]: Server ready
Mar 29 18:53:07 sufflain.icu systemd[1]: Reloading Fail2Ban Service.
Mar 29 18:53:07 sufflain.icu fail2ban-client[450737]: OK
Mar 29 18:53:07 sufflain.icu systemd[1]: Reloaded Fail2Ban Service.
~
~
~
~
lines 1-17/17 (END)
```

Docker



Install Docker Engine | Docker Docs

https://docs.docker.com/engine/install/

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Manuals / Docker Engine / Install

Overview

Install

Installation per distro

Optional post-installation steps

Release notes

Previous versions

Deprecated features

Docker Buildx

Docker Context

Docker Scan

Docker Compose

Docker Hub

Docker subscription

Administration

Security

Open-source projects

Desktop

Platform	x86_64 / amd64	arm64 (Apple Silicon)
Docker Desktop for Mac (macOS)	✓	✓
Docker Desktop for Windows	✓	

Server

Docker provides `.deb` and `.rpm` packages from the following Linux distributions and architectures:

Platform	x86_64 / amd64	arm64 / aarch64	arm (32-bit)	s390x
CentOS	✓	✓		
Debian	✓	✓	✓	
Fedora	✓	✓		
Raspbian			✓	
RHEL				✓
SLES				✓

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Request docs changes

On this page:

Supported platforms

Desktop

Server

Other Linux distributions

Release channels

Stable

Test

Nightly

Support

Backporting

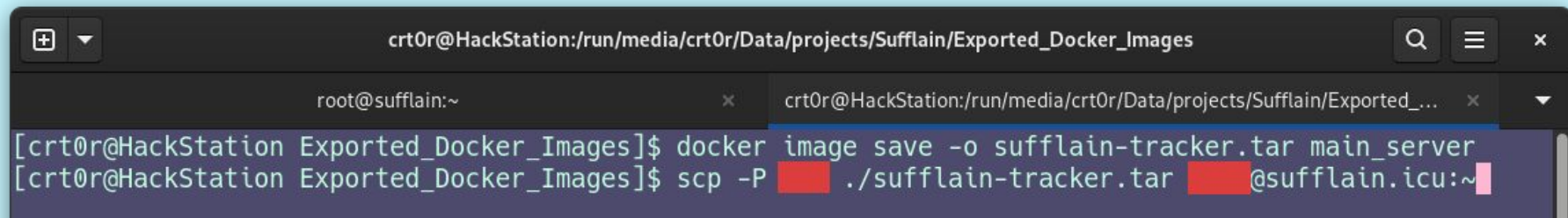
Upgrade path

Licensing

Reporting security issues

Get started

Сохранение и выгрузка образа



A terminal window with a dark theme. The title bar shows the path `crt0r@HackStation:/run/media/crt0r/Data/projects/Sufflain/Exported_Docker_Images`. There are two tabs: `root@sufflain:~` and `crt0r@HackStation:/run/media/crt0r/Data/projects/Sufflain/Exported_...`. The active tab shows the following commands and their prompts:

```
[crt0r@HackStation Exported_Docker_Images]$ docker image save -o sufflain-tracker.tar main_server  
[crt0r@HackStation Exported_Docker_Images]$ scp -P [REDACTED] ./sufflain-tracker.tar [REDACTED]@sufflain.icu:~
```

Загрузка и развертывание

```
root@sufflain:~  
> docker load -i /home/[REDACTED]/sufflain-tracker.tar
```

```
root@sufflain:~  
> docker run -it -d --name sfl --restart unless-stopped main_server
```

```
root@sufflain:~  
> docker run -d -p 8080:80 --name www --restart unless-stopped sufflain-web-server
```

**Решения по улучшению инфраструктуры и
анализ предпринятых мер безопасности**

Activities Google Chrome Mar 29 19:06 en 38 %

106.53.60.245 | Tencent Cloud Computing (Beijing) Co. Ltd. | AbuseIPDB - Google Chrome

fail2ban - Google x Fail2ban - Wikip x In Fail2Ban, How x selinux-ssh what x Fail2ban Loggin x tmux remove sel x vim - Selecting te x bash - How to st x 106.53.60.245 | x IP Info: 41.42.112 x +

abuseipdb.com/check/106.53.60.245

 **AbuseIPDB**

Home Report IP Bulk Reporter Pricing About FAQ Documentation Statistics IP Tools Contact LOGIN SIGN UP

AbuseIPDB » 106.53.60.245

Check an IP Address, Domain Name, or Subnet
e.g. 185.77.207.46, microsoft.com, or 5.188.10.0/24

185.77.207.46 CHECK

106.53.60.245 was found in our database!

This IP was reported **437** times. Confidence of Abuse is **100%** ?

100%

ISP Tencent Cloud Computing (Beijing) Co. Ltd.

Usage Type Data Center/Web Hosting/Transit

Domain Name tencent.com

Country  China

City Beijing, Beijing

IP Info including ISP, Usage Type, and Location provided by [IP2Location](#).
Updated monthly.

REPORT 106.53.60.245 WHOIS 106.53.60.245

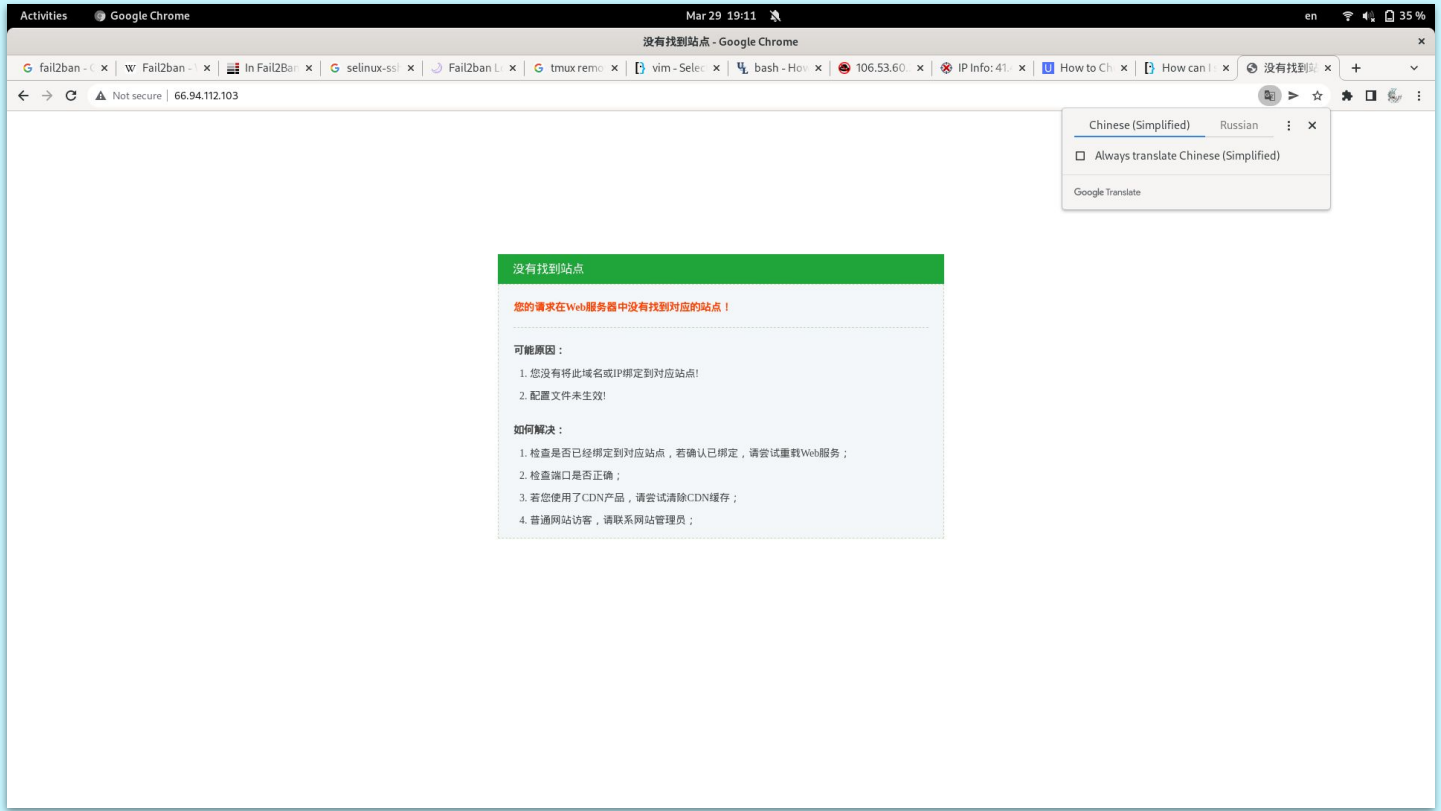
 Students and Teachers, save up to 60% on Adobe Creative Cloud.

ADS VIA CARBON

IP Abuse Reports for 106.53.60.245:

This IP address has been reported a total of **437** times from 49 distinct sources. 106.53.60.245 was first reported on February 7th 2022, and the most recent report was **17 minutes ago**.

feedback





mobile operator injects ads into http page



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[Shopping](#)

[More](#)

[Tools](#)

About 179,000,000 results (0.48 seconds)

<https://www.pcworld.com> > Security > News

AT&T Wi-Fi hotspot caught injecting ads into web pages

Aug 26, 2015 — AT&T is the latest Wi-Fi hotspot provider to try injecting ads into sites where they shouldn't be.

<https://thehackernews.com> > computer security > hacking

Chinese ISPs Caught Injecting Ads and Malware into Web ...

Feb 27, 2016 — Here's How Malware and Ads are Injected. In the research paper titled 'Website-Targeted False Content Injection by Network Operators,' the ...

<https://www.csoonline.com> > Browsers

T-Mobile caught in crossfire of war over injected ads - CSO ...

Aug 3, 2015 — LAS VEGAS - An ongoing conflict between website owners and ad injectors who place unwanted ads on those websites has just flared up into full- ...

<https://www.computerworld.com> > Malware

Wikipedia warns users about malware injecting ads into its ...

May 15, 2012 — Rogue browser extensions might inject commercial ads into Wikipedia pages, Wikimedia Foundation said.

<https://www.theverge.com> > hotel-caught-injecting-adv...

Hotel caught injecting advertising into webpages ... - The Verge

Apr 7, 2012 — Arbitrary HTML page rewriting enables the operator to inject static or dynamic advertising onto every web page the end-user views.

<https://www.usenix.org> > sec16_paper_nakibly PDF

Website-Targeted False Content Injection by Network Operators

by G Nakibly · 2016 · Cited by 29 — primary focus of these network operators is to inject advertisements into web pages, we also identified injections of malicious content.

19 pages

Добавление обратного прокси

```
7 server {  
8     server_name sufflain.icu;  
9  
10    location / {  
11        proxy_pass http://127.0.0.1:8080;  
12    }
```

```
Activities Terminal Mar 12 10:51 en 49 %
root@sufflain:/etc/nginx

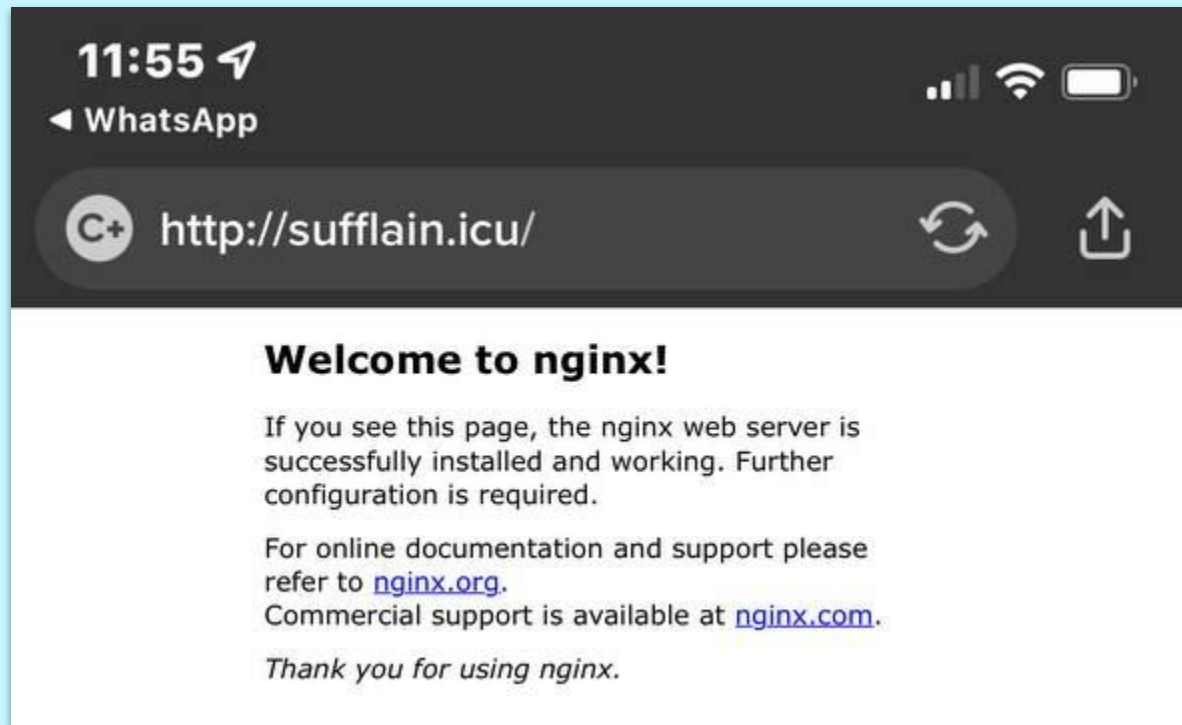
> certbot --nginx -d sufflain.icu -d www.sufflain.icu /etc/nginx@sufflain.icu
Saving debug log to /var/log/letsencrypt/letsencrypt.log
Plugins selected: Authenticator nginx, Installer nginx
Enter email address (used for urgent renewal and security notices) (Enter 'c' to
cancel): crt0r.9@yahoo.com

-----
Please read the Terms of Service at
https://letsencrypt.org/documents/LE-SA-v1.2-November-15-2017.pdf. You must
agree in order to register with the ACME server at
https://acme-v02.api.letsencrypt.org/directory
-----
(A)gree/(C)ancel: A

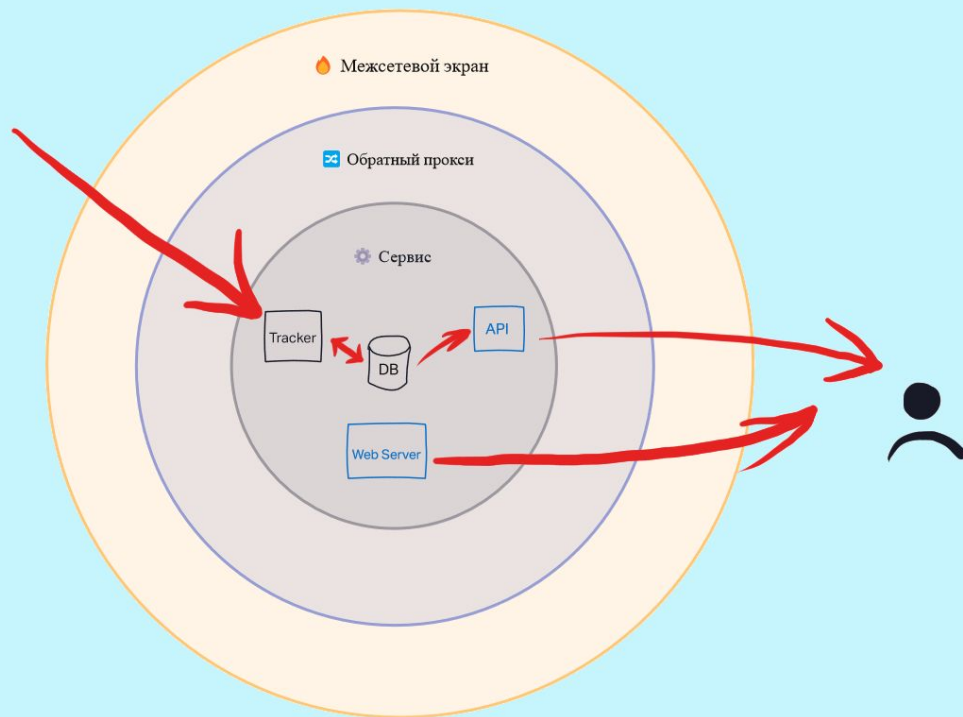
-----
Would you be willing to share your email address with the Electronic Frontier
Foundation, a founding partner of the Let's Encrypt project and the non-profit
organization that develops Certbot? We'd like to send you email about our work
encrypting the web, EFF news, campaigns, and ways to support digital freedom.
-----
(Y)es/(N)o: Y
Obtaining a new certificate
Performing the following challenges:
http-01 challenge for sufflain.icu
http-01 challenge for www.sufflain.icu
Waiting for verification...
Cleaning up challenges
Deploying Certificate to VirtualHost /etc/nginx/sites-enabled/default
Deploying Certificate to VirtualHost /etc/nginx/sites-enabled/default

Please choose whether or not to redirect HTTP traffic to HTTPS, removing HTTP access.
-----
1: No redirect - Make no further changes to the webserver configuration.
2: Redirect - Make all requests redirect to secure HTTPS access. Choose this for
new sites, or if you're confident your site works on HTTPS. You can undo this
change by editing your web server's configuration.
```

Проблемы после обновления Nginx



Новая архитектура



Спасибо за внимание

